

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – INTERACTIVE TREE SIMULATOR

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	Assessment:	Assessment Tool 1 – Interactive Tree Simulator
Weightage:	30% of CIA	Total Marks:	100
CO3 Statement:	<i>Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, B-Trees, Red-Black Trees, and Splay Trees.</i>		
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Section / Batch:		Date:	27/07/2026

Code of Conduct

I _____ (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: _____

k.kaviya

Instructions

- This assessment tool contains 5 Interactive Tree Simulator questions, Total: 100 Marks.
- Students can use any online/offline Interactive Tree Simulator. Demonstrate insertion, deletion, searching, and traversal operations wherever applicable.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation.
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Section / Topic	Focus	Q Nos	Marks
Binary Search Tree	Properties, binary tree and binary search tree, insertion and deletion	1	20
Tree Traversals	Traversal types, In order, Post order and Pre order	2	20
AVL Tree	Balancing factor, Rotation, Types of rotation, examples	3	20
B Tree, 2-3 tree, 2-3-4 tree	Properties, structures and Operations	4	20
Red Black Tree	Properties, Example and Operations	5	20
GRAND TOTAL:			100 Marks

36.

37.

38.

39.

1. Construct a Binary Search Tree (BST) by inserting the following keys:
40, 20, 10, 30, 60, 50, 70

Perform:

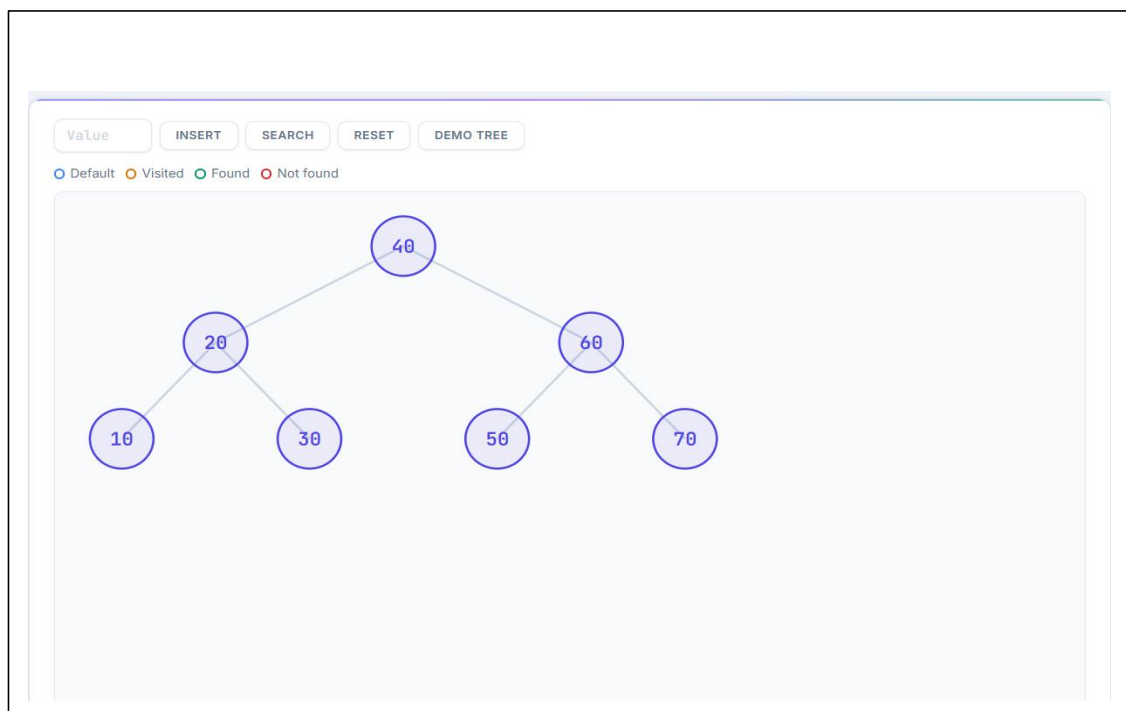
a. In order Traversal

b. Search for 70

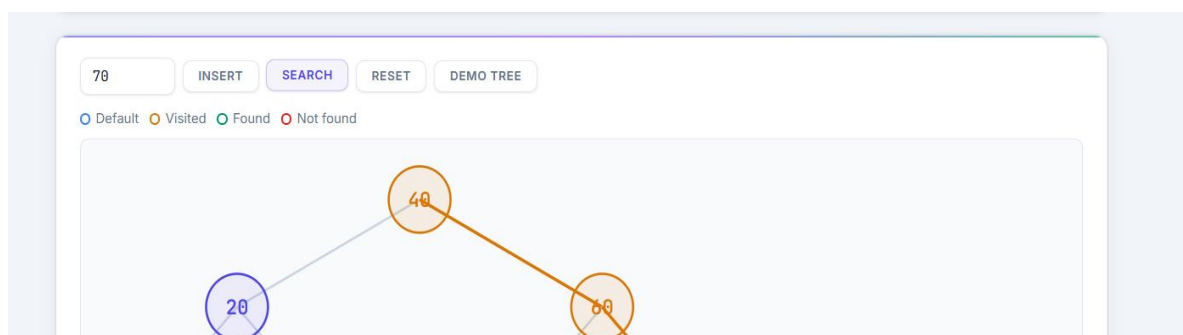
c. Delete 50

d. Display the final BST.

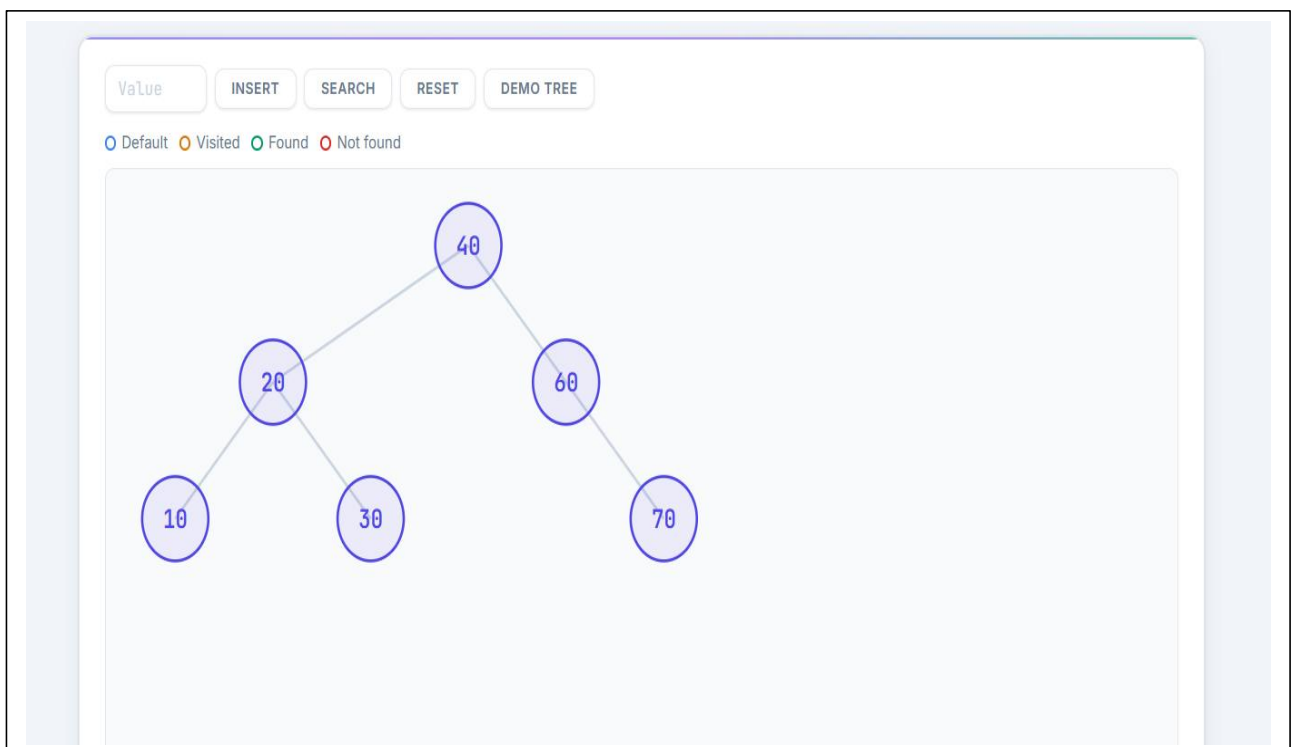
a. In order Traversal



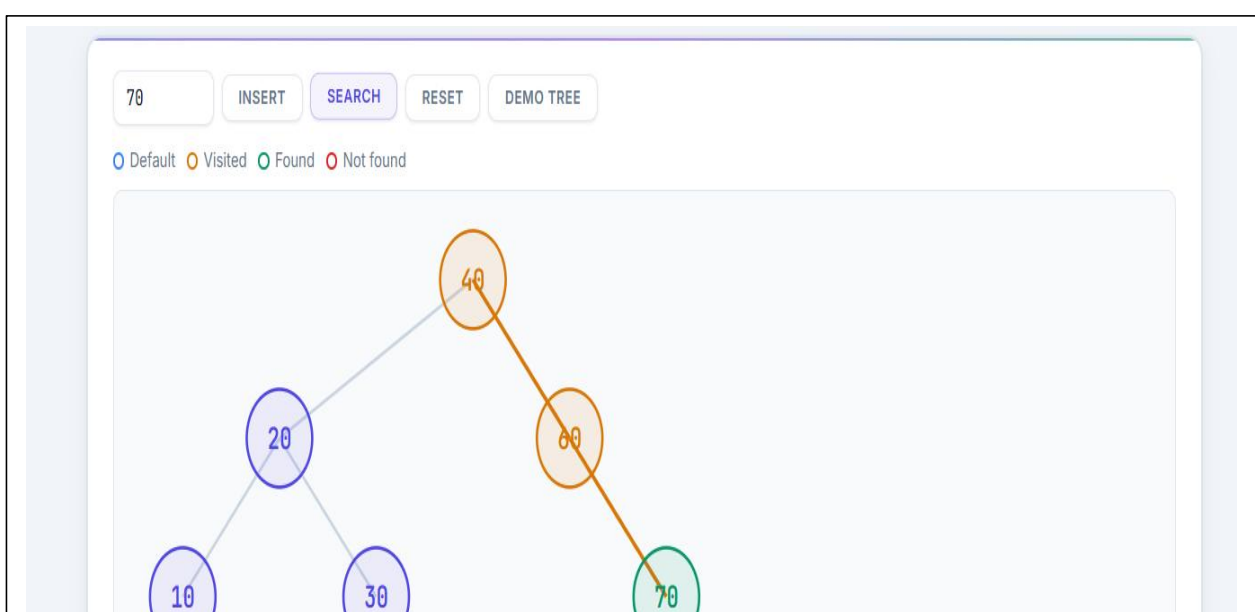
b. Search for 70



c. Delete 50



d. Display the final BST



2. Complete Tree Traversals by Inserting 55,35,75,25,45,65,85. Find all the traversals.

a. Inorder

b. Preorder

c. Postorder

a. Inorder

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Tree Traversals

Input

Input is level-by-level order.

55,35,75,25,45,65,85

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

25354555657585

55

35

75

25

45

65

85

b. Preorder

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Tree Traversals

55 35 25 45 75 65 85

Input

Input is level-by-level order.

55,35,75,25,45,65,85

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

c. Postorder

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Tree Traversals

Input

Input is level-by-level order.

55,35,75,25,45,65,85

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

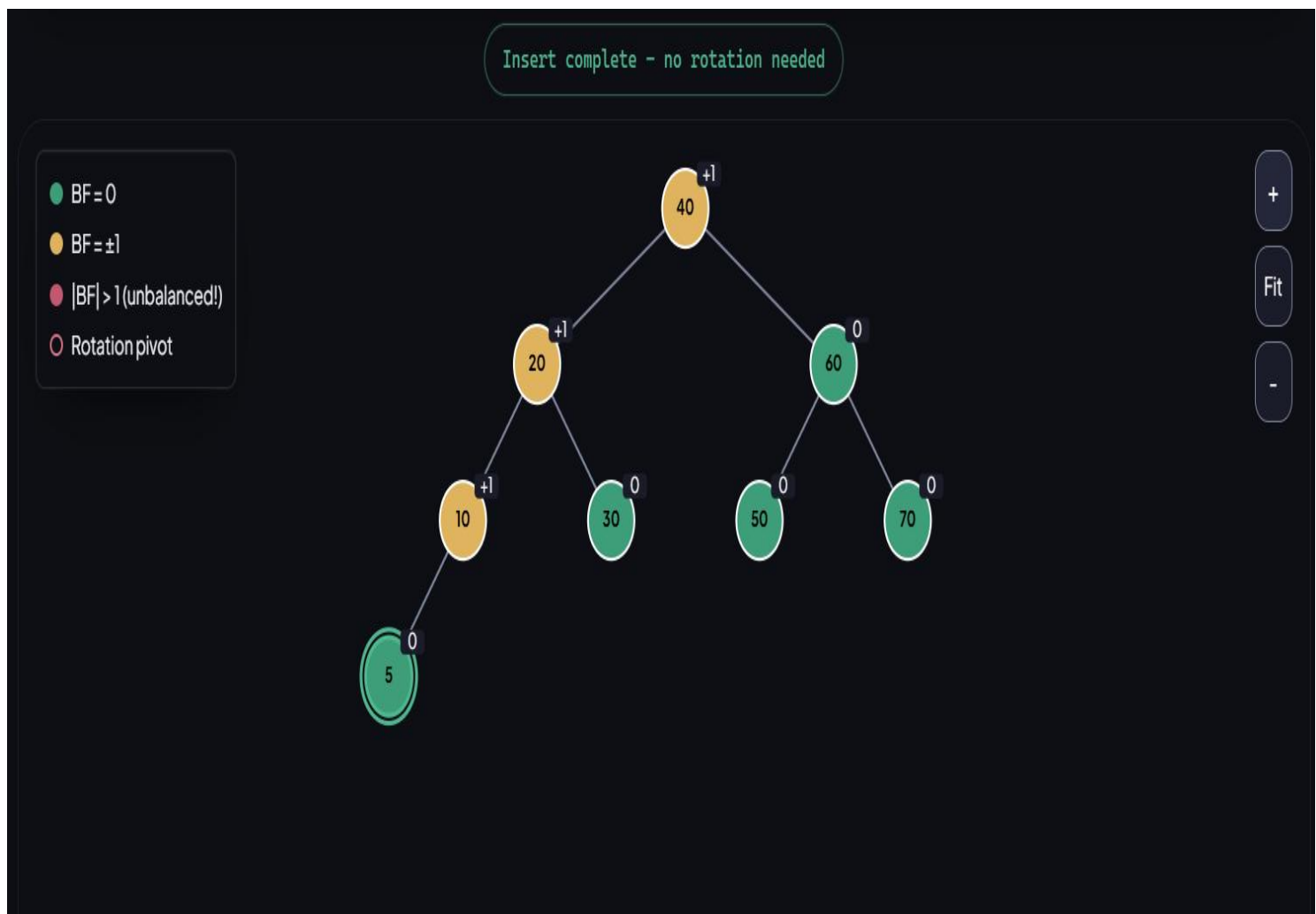
Start Simulation

25 45 35 65 85 75 55

```

graph TD
    55((55)) --- 35((35))
    55 --- 75((75))
    35 --- 25((25))
    35 --- 45((45))
    75 --- 65((65))
    75 --- 85((85))
  
```

3. Insert 40, 20, 60, 10, 30, 50, 70, 5 into an AVL Tree. Determine the balance factor of every node before and after balancing.



4. Construct a B-Tree of order 3 by inserting 10, 20, 30, 40, 50.

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 **M-ary Tree**

M Value (Max children per node)

3

Choose between 2 and 10

Input

Input is level-by-level order for M-ary tree.

10,20,30,40,50

Enter numbers separated by commas. Use 'null' for empty nodes.

Visualize

Generate Python Code



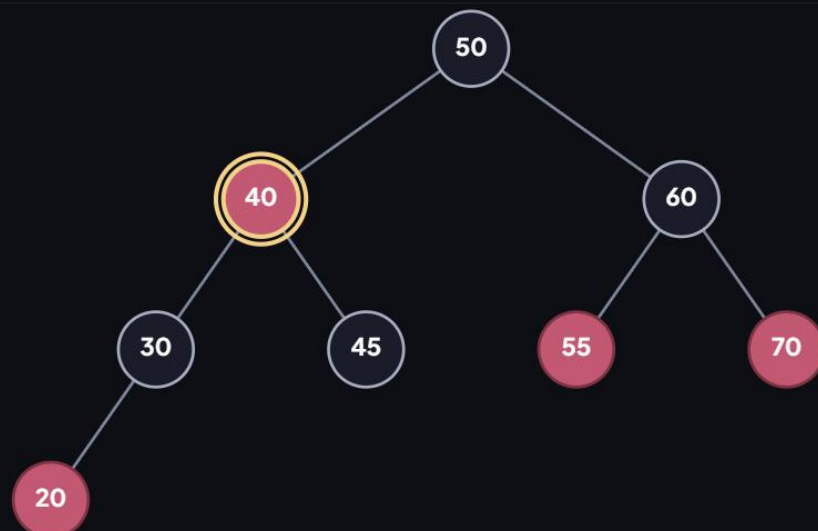
```
graph TD; 10((10)) --- 20((20)); 10 --- 30((30)); 10 --- 40((40)); 20 --- 50((50));
```

5. Construct a Red-Black Tree using 50, 40, 60, 30, 45, 55, 70, 20. Delete 30 and analyze how Red-Black properties are restored.

Case 1: Uncle Red → Recolor Parent, Uncle & GP

PROPERTIES (LIVE)

- ✓ Root is black
 - ✓ No red-red parent/child
 - ✓ Equal black heights
- Black height: 3



Delete 30

Fixup Complete. Current black.

PROPERTIES (LIVE)

- ✓ Root is black
 - ✓ No red-red parent/child
 - ✓ Equal black heights
- Black height: 3

